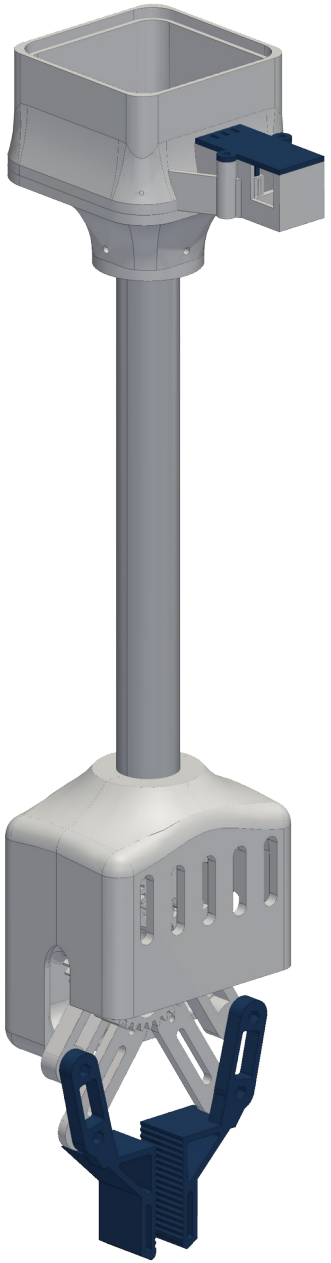
RoomCleaner

A four-cable ceiling robot that picks laundry off the floor

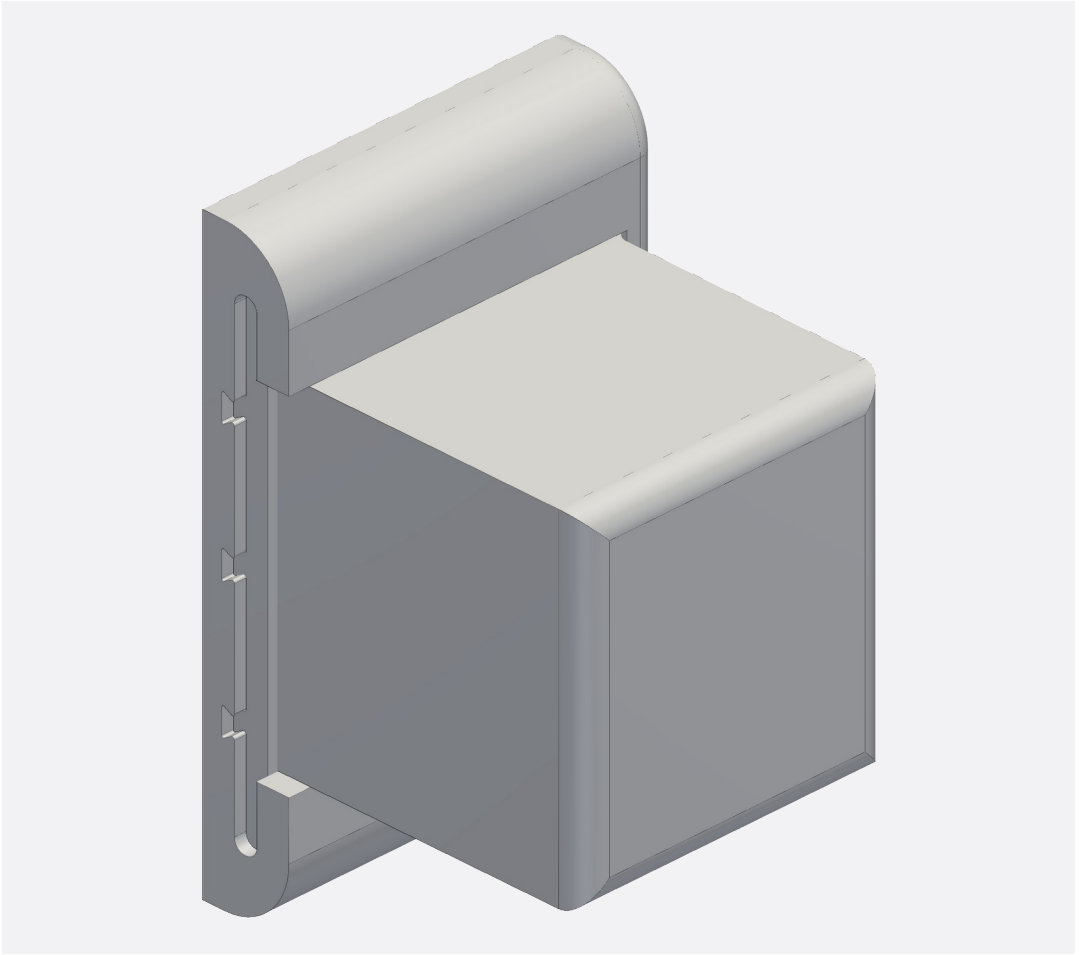
Jasper Buntinx
Technical presentation · October 27, 2026





Goal and constraints

- Pick clothes off the floor and drop them in a hamper, from anywhere in the room
- Four winches near the ceiling corners hold one claw on four cables
- Room 11 × 9 ft with a 9 ft ceiling
- Working load: a pair of jeans (0.9 kg) with 2 m/s^2 of acceleration; first test load 1 lb
- 3D-printed in PETG on a hobby budget, from parts I can buy and fix myself
- Safety: anchors in studs, one hard stop (the switched power strip), nobody under the robot



One of the four winch stations (CAD)

Outline

1 **How a cable robot works**
Position vectors, statics, the winch

2 **From one box to modular stations**
Wiring, a controller per winch, the slide-in housing, plug-in power, cooling and the horn-inspired fin

3 **Homing and calibration**
Bead homing, why it limited the motion, the camera-based method

4 **The claw**
Gripper evolution, the four-bar hands, overhead camera to claw camera, weight vs lift

5 **Manufacturing**
Printed PETG parts, and a heat fin cut and drilled on a manual mill

6 **Electronics and software**
Station board, bring-up, control and the phone app

7 **Status and next steps**
First test, what is built, what I learned

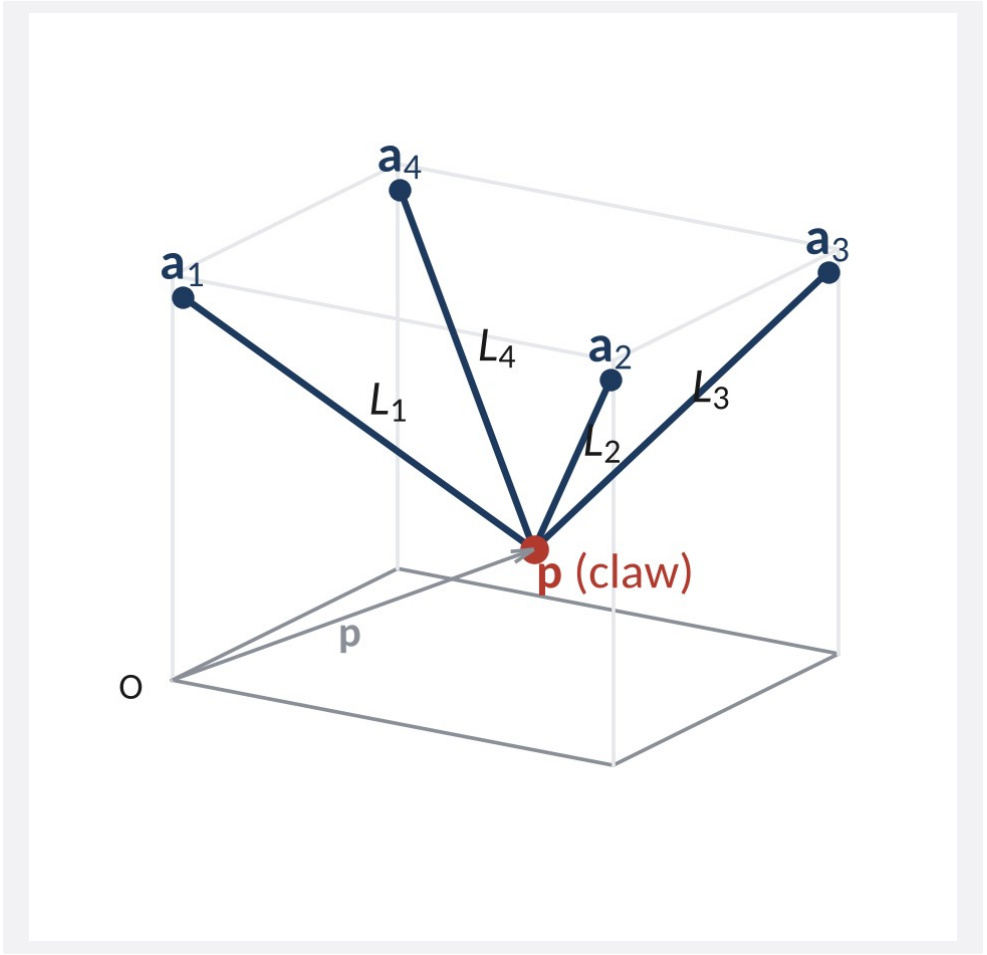
01

How a cable robot works

Four cables, one point, and the vectors that connect them

Position vectors

- Room frame at a floor corner: anchor i (the pulley exit) at $\mathbf{a}_i$, the claw at $\mathbf{p}$
- Each cable is the vector from the claw to its anchor
- Its length is the distance between those two points
- To move, compute the four lengths at every point on the path
- Each spool turns by the change in length over its radius
- Closed form: no iteration is needed to command a move



EQUATIONS

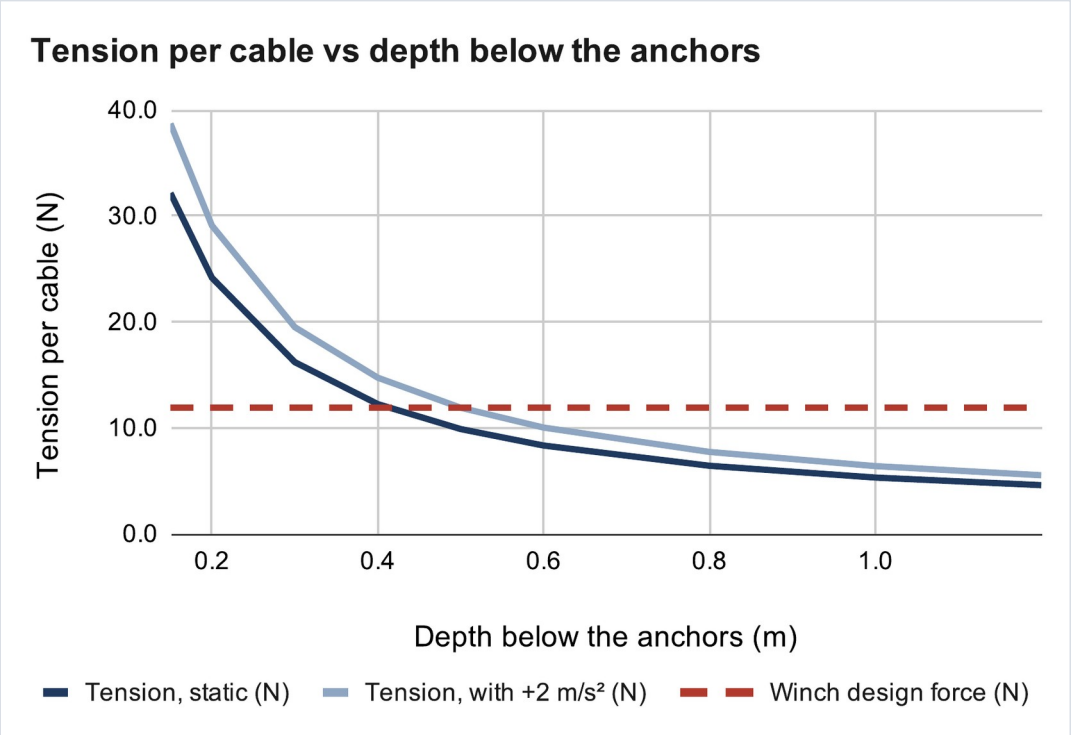
$$\mathbf{l}_i = \mathbf{a}_i - \mathbf{p}, \quad \mathbf{u}_i = \mathbf{l}_i / L_i$$

$$L_i = \|\mathbf{a}_i - \mathbf{p}\| = \sqrt{((a_{ix} - p_x)^2 + (a_{iy} - p_y)^2 + (a_{iz} - p_z)^2)}$$

$$\Delta\theta_i = \Delta L_i / r \quad (r = \text{spool radius with the wound line})$$

Statics: why cable angle matters

- Force balance: the cable pulls and the weight sum to zero
- Cables only pull: every tension must stay positive
- Centred claw, equal shares: tension rises as the cables flatten
- 32 N per cable 0.15 m below the anchors; 8.3 N at 0.6 m
- At 0.4 m: 12.2 N, right at the 11.9 N design force
- So the robot keeps 0.4 m below the anchors, 0.2 m from the walls



Expected values: my calculation sheet (Google Sheets), "Cable tension" tab

EQUATIONS

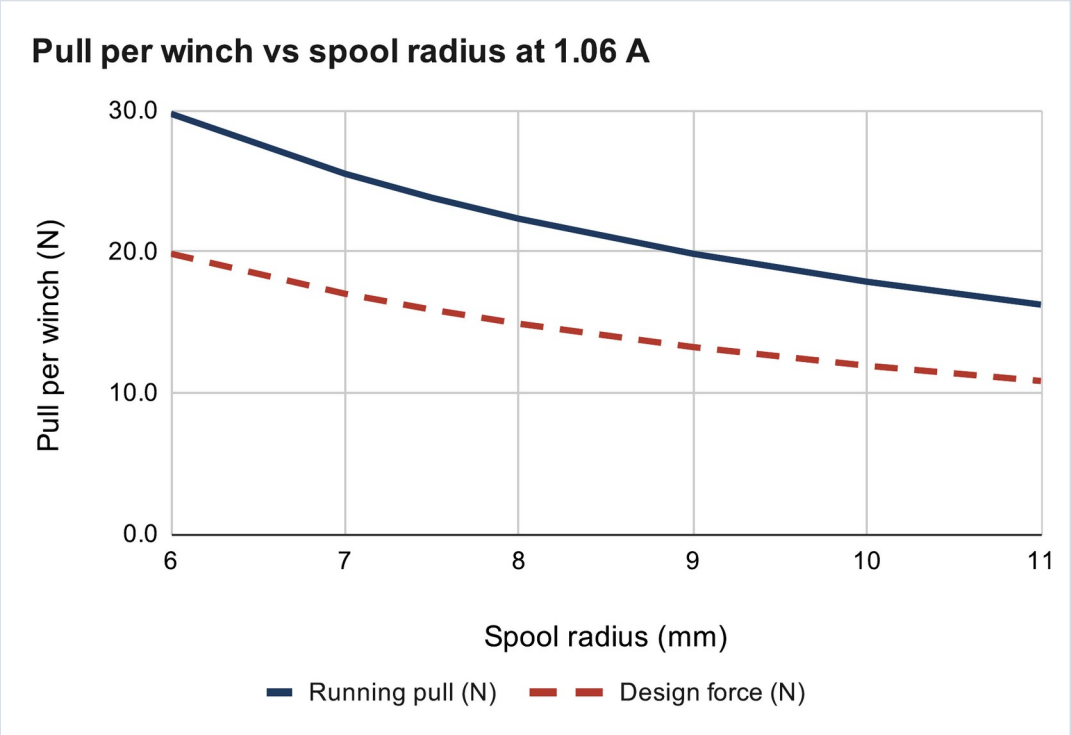
$$\Sigma\, T_i\, \mathbf{u}_i + m(\mathbf{g} + \mathbf{a}) = \mathbf{0}, \quad T_i > 0$$

$$T = W / (4 \sin\varphi), \quad \sin\varphi = d / \sqrt{(d^2 + D_h{}^2)}$$

$$W = (0.45 + 0.454)\ \text{kg} \times 9.81\ \text{m/s}^2 = 8.87\ \text{N}$$

The winch

- NEMA 17 stepper, printed spool, outlet pulley
- Current limit set by the driver's reference voltage: 0.85 V → 1.06 A
- Running torque < holding torque: 0.7 assumed, the pull test measures it
- Smaller spool: more pull per amp, more turns per metre
- v3 spool: 24 N empty, 18 N full → 11.9 N design (÷ 1.5)



Expected values: my calculation sheet (Google Sheets), "Winch pull" tab

EQUATIONS

$$I = V_{ref} / (8\ R_s) = 0.85\ V / (8 \times 0.1\ \Omega) = 1.06\ A$$

$$\tau_{run} = 0.7 \cdot \tau_{hold} \cdot I / I_{rated} = 0.7 \times 0.36 \times 1.06 / 1.5 = 0.18\ N{\cdot}m$$

$$T = \tau_{run} / r: \quad 23.8\ N \text{ at } r = 7.5\ mm, \quad 17.9\ N \text{ at } r = 10\ mm$$

Current limit: Allegro MicroSystems, A4988 datasheet

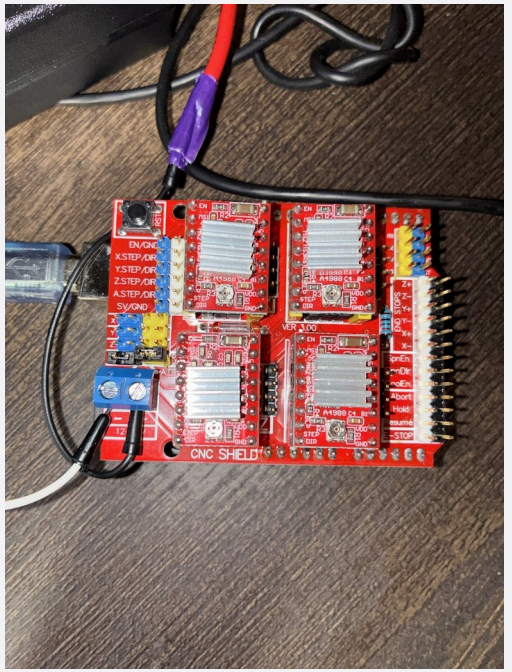
02

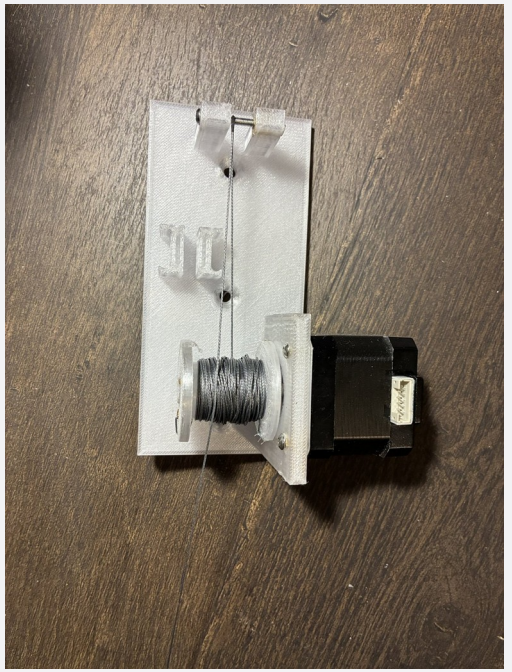
From one box to modular stations

How adding safety features changed the whole architecture

First idea: one central controller

- One Arduino Uno and a CNC shield with four A4988 drivers, in a single box
- The computer sends moves; the shield steps all four motors from one clock, so sync is free
- Each motor gets a long 4-wire cable from the box to its corner
- It worked on the bench: a gen-1 station lifted a test weight

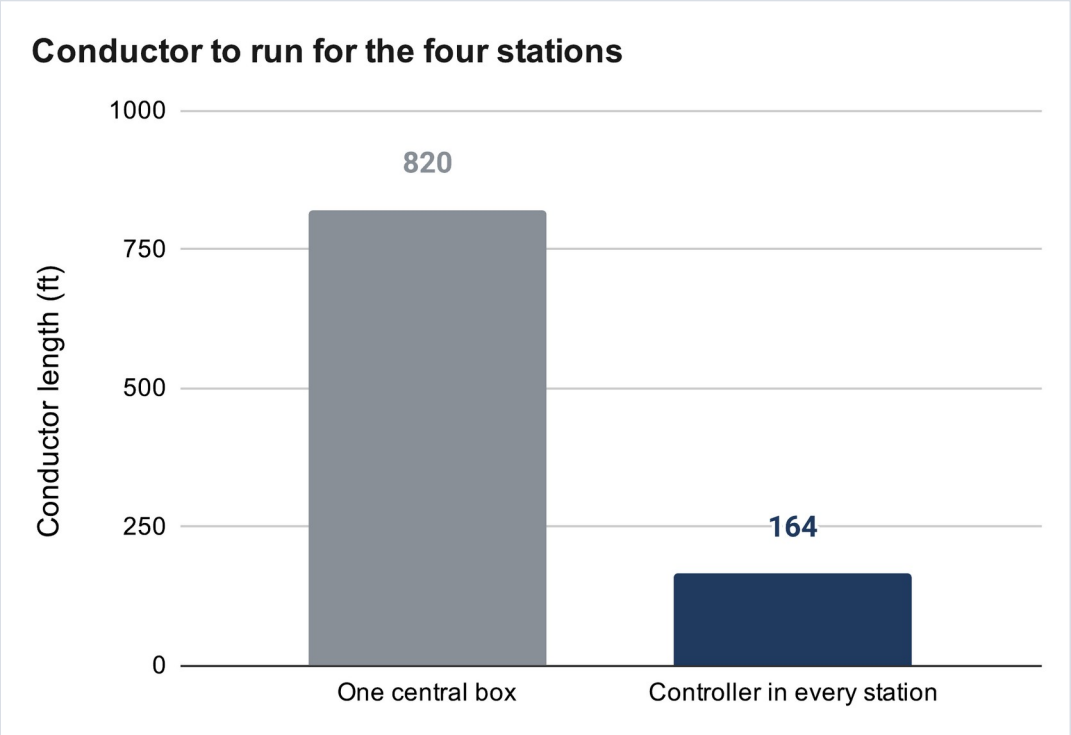




Gen 1: central shield with four drivers; spool plate on the wall

Why it did not scale: the wiring

- I wanted more safety: a homing switch and an encoder at every winch, plus fault detection
- Per corner: motor 4 + switch 2 + encoder 4 = **10 conductors**, not 4
- Real cable routes are 6, 21, 23 and 32 ft (82 ft)
- The encoder's I²C bus is meant for centimetres, not 10 m runs
- Motor phase currents in long cables also radiate noise into the sensor lines



Expected values: my calculation sheet (Google Sheets), "Wiring" tab

EQUATIONS

$$\text{Conductor} = n \cdot \Sigma L: \quad 10 \times 82\ \text{ft} = 820\ \text{ft} \quad \text{vs} \quad 2 \times 82\ \text{ft} = 164\ \text{ft}$$

$$\Delta V = I \cdot R' \cdot 2L = 2\ \text{A} \times 4.016\ \text{m}\Omega/\text{ft} \times 64\ \text{ft} = 0.51\ \text{V}$$

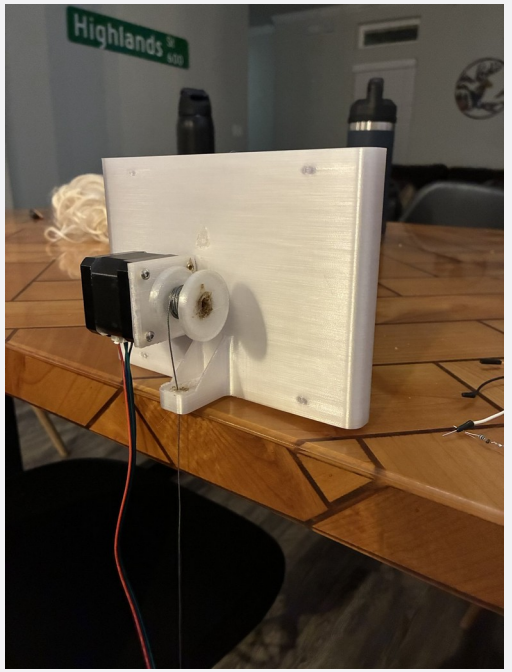
16 AWG on the longest run (32 ft); 2 A is illustrative

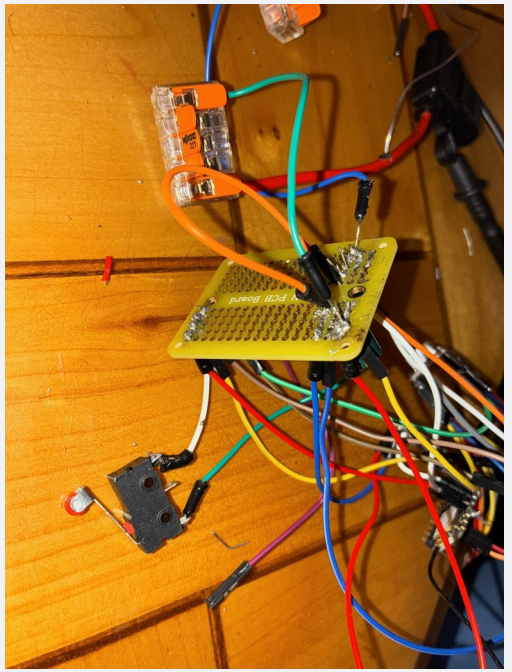
Decision: a controller in every winch

- Each station: XIAO ESP32-C3 + A4988 driver + 5 V buck; only 12 V power travels, on 2 wires
- Switch and encoder wires stay inside the station, a few centimetres long
- The host computer plans the path and sends each station timed segments over Wi-Fi
- Each station runs its segments on its own clock; any lost link → hold position
- Cost: timing skew between stations becomes cable mismatch, so segments are time-stamped
- Benefit: about 5× less wire, and every station can be built and tested alone

EQUATIONS

$$\Delta x = v \cdot \Delta t = 50\ \text{mm/s} \times 50\ \text{ms} = 2.5\ \text{mm of cable mismatch}$$

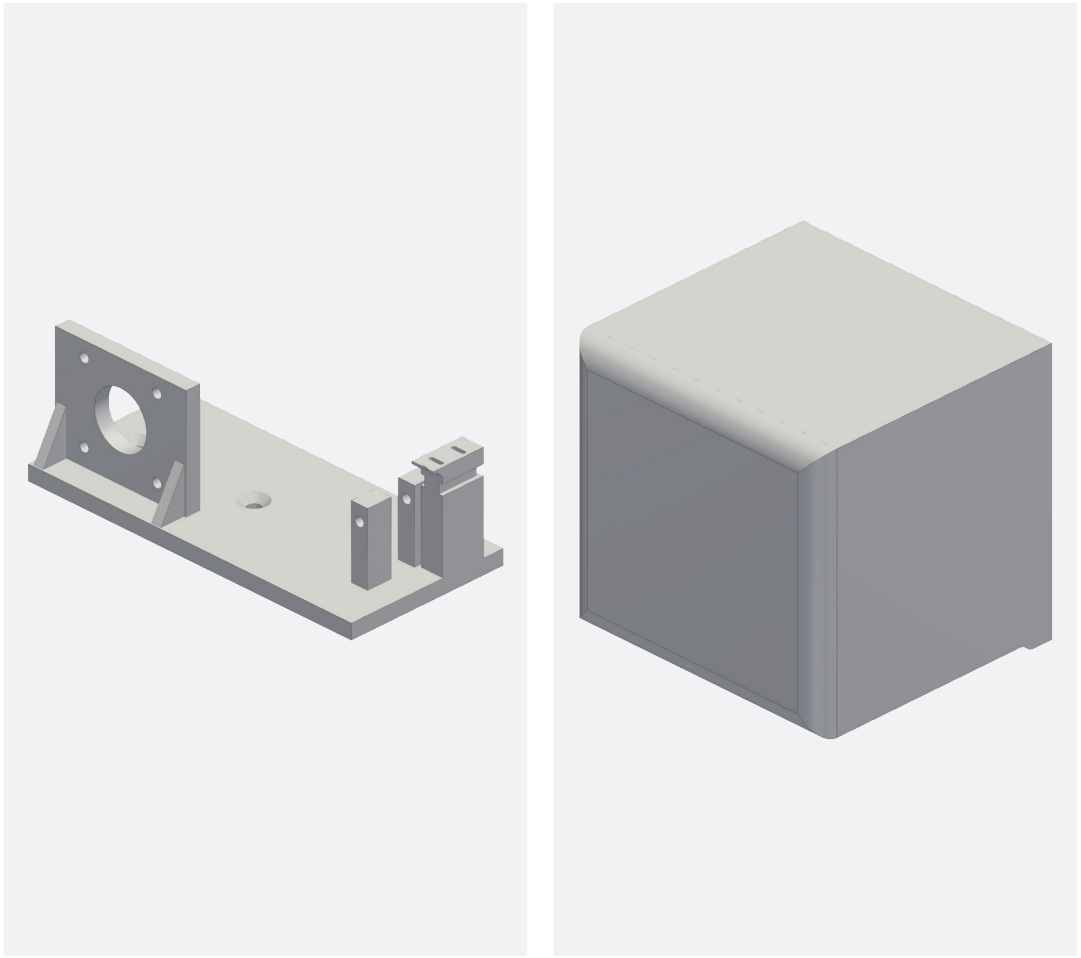




Gen 2: the station carries its own controller, driver and buck

Making the units modular

- Gen 1 was a plate screwed into the wall: fine for a bare motor
- Each corner now carries a controller, driver, buck, switch and encoder
  - they need protection, a clean look and a path for heat
- So each station became a housing: motor, spool, pulley and electronics in one unit
- Same unit in all four corners: build one, test one, copy it



Gen-1 corner plate (left) vs the gen-2 station housing (right)

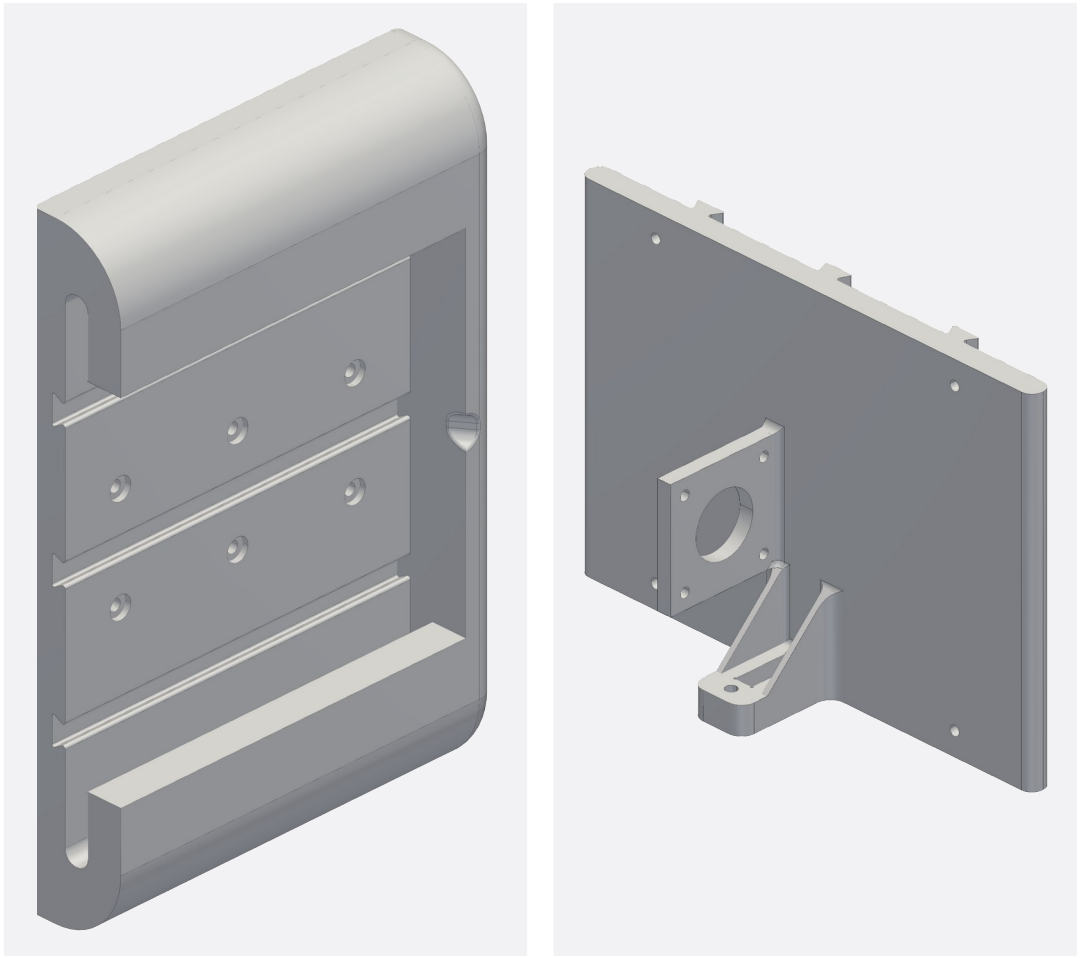
Slide-in mount

- I take stations down often: to test, rework, repair or modify them
- Drilling in and out each time damages the wall and moves the anchor point
- A fixed wall frame goes into a stud once; the station slides in and locks
- Pull it out, rework it, slide it back: the frame keeps the anchor position
- Load: the line pulls the station away from the wall (pull-out and prying)
  - design target: SF 3 at 34 N; SolidWorks study planned

EQUATIONS

$$SF = \sigma_{allow} / \sigma_{max} \geq 3 \rightarrow \sigma_{max} \leq 25\ MPa / 3 = 8.3\ MPa$$

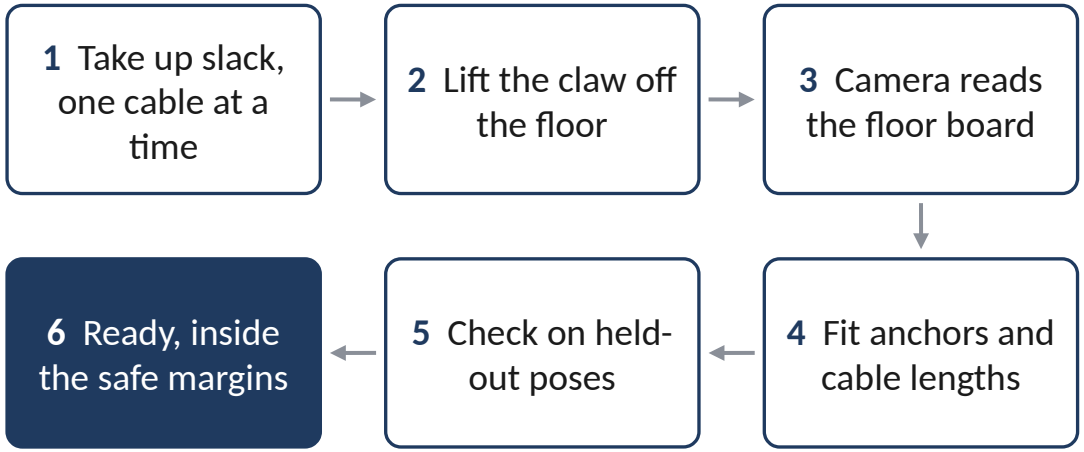
Allowable: derated PETG yield, 25 MPa

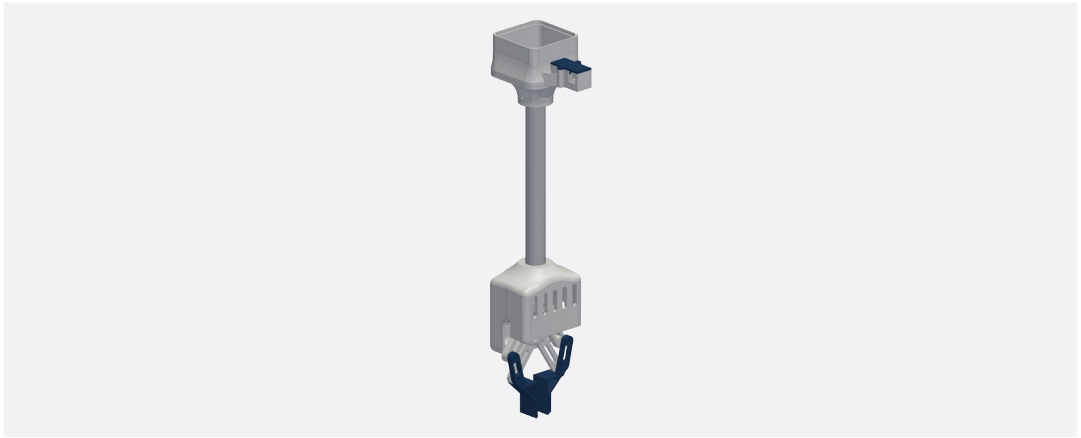


Fixed wall frame (left) and the slide-in piece (right)

Self-calibration: set it up anywhere

- Every time the stations go back up, the robot measures its own geometry
  - no tape measure, no numbers typed in, nothing for a user to get wrong
- The claw camera reads a printed floor board (a grid of tags) to get the claw's pose
- The robot moves through a set of poses and fits the anchor positions and cable lengths by least squares
- Works in any room the cables can reach, so it can become a consumer product

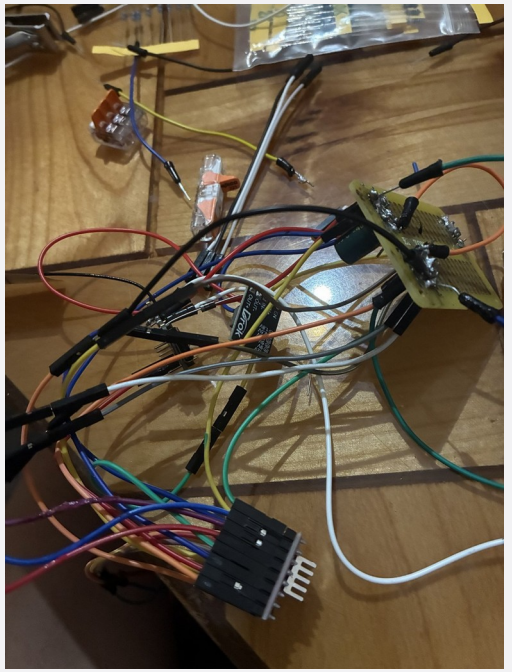


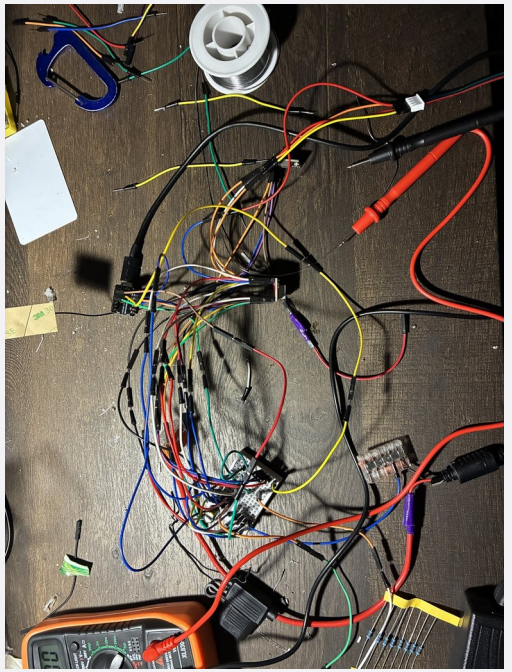


Startup calibration; the claw camera looks straight down at the board

Plug-in power

- Each station has one cable: a 16 AWG pair from a fused 12 V supply
- Plug in to use; unplug and store the cables when the robot is not needed
- Stations slide off the wall frames if wanted
- Runs cut at 9, 24, 26 and 35 ft (94 ft of a 100 ft roll), a fuse per branch at the supply
- The motor stays wired inside the station, so it is never unplugged under power
- The switched power strip is the one hard stop

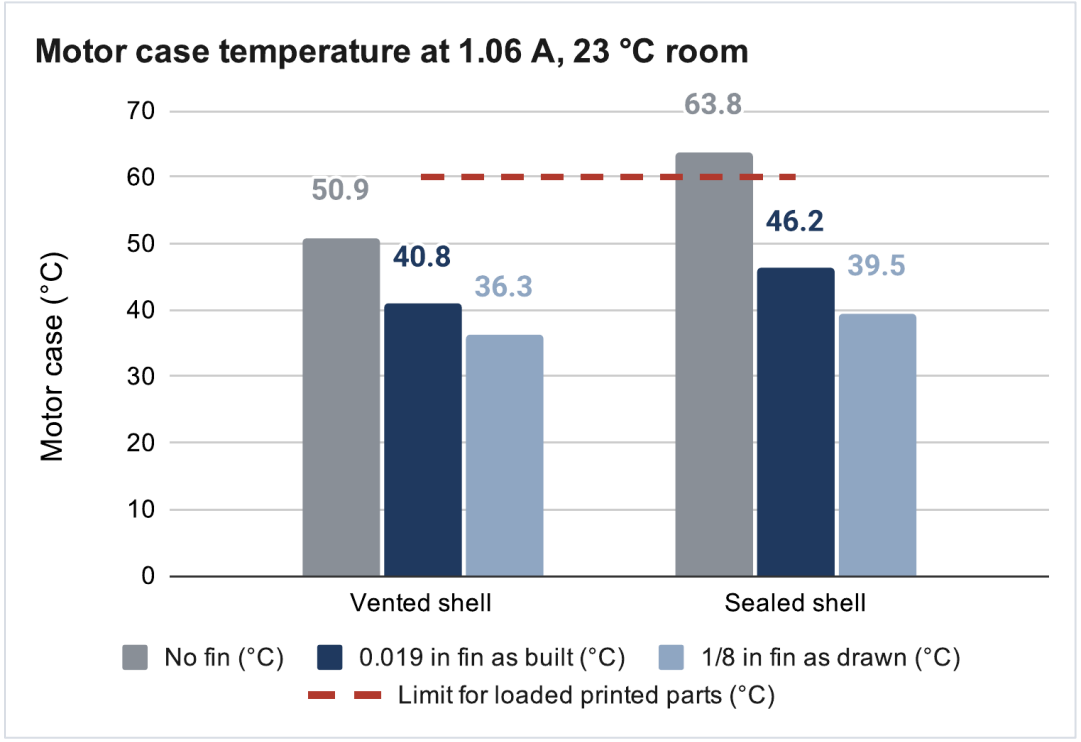




Inside a station: the buck converter and its wiring; only 12 V comes in

Cooling

- A sealed printed housing traps the motor's heat
  - PETG at the loaded motor mount softens and creeps: 60 °C limit
- Fix: vents, plus an aluminium fin straight up through the shell
- Model: motor → shell air → room, the fin a second path out
- At 1.06 A: sealed 64 °C without the fin, 46 °C with it
- A 90-min hold test will measure it



Expected values: my calculation sheet (Google Sheets), "Motor temperature" tab

EQUATIONS

$$P_m = I^2 R = (1.06\ \text{A})^2 \times 1.8\ \Omega = 2.0\ \text{W}$$

$$\Delta T = q \cdot R_{th} \quad \text{(heat flow} \times \text{thermal resistance)}$$

$$\text{Fin in parallel with the shell: } 1/R = 1/R_{shell} + 1/R_{fin}$$

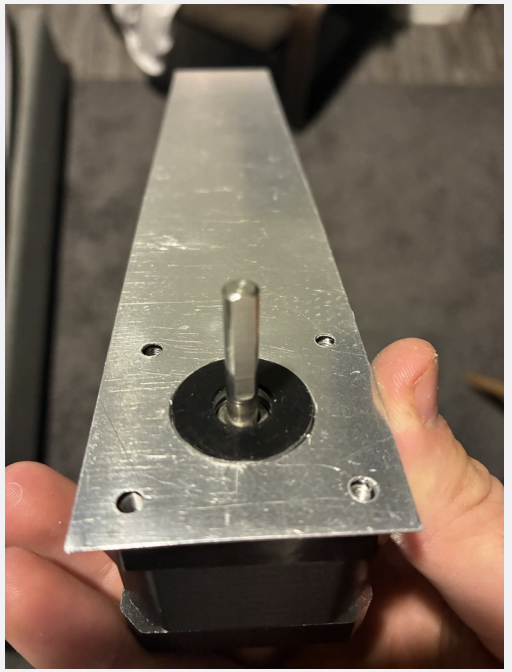
Bio-inspired fin: antelope horns

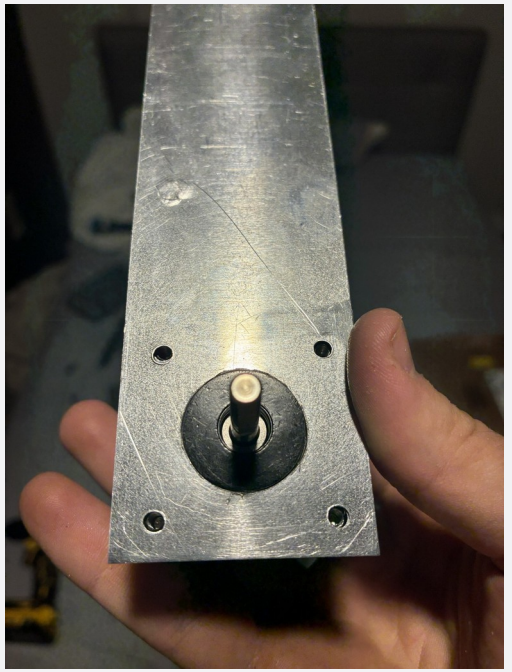
- Idea from my bio-inspired design class: antelope horns
- A horn is a heat exchanger: warm blood flows into its core, heat leaves through the surface
- The fin copies that: one tall, thin blade rising straight up from the heat source
  - heat conducts up from the motor and leaves along the length; the tip runs cooler, like a horn's
- Vertical, so warm air rises freely off both faces
- Motor width (44 mm): adds height, not width, to the station

EQUATIONS

Horn and fin both shed heat by convection: $q = h \cdot A \cdot (T_s - T_\infty)$

Tall and thin: large area for little mass, $A = 2\,w\,L$





The fin on the motor: one tall blade rising from the heat source

Heat fin: how much length is useful

- The far end runs cooler, so each extra mm does less
- **0.019 in sheet (built):** 95 % of a long fin's heat by **136 mm**
- **1/8 in plate (drawn):** 95 % by **349 mm**
- Built 200 mm above the cover: past 136 mm the sheet adds < 5 %
- Looks: horn-like, straight up at motor width (44 mm), matte black
  - black radiates more than bare aluminium

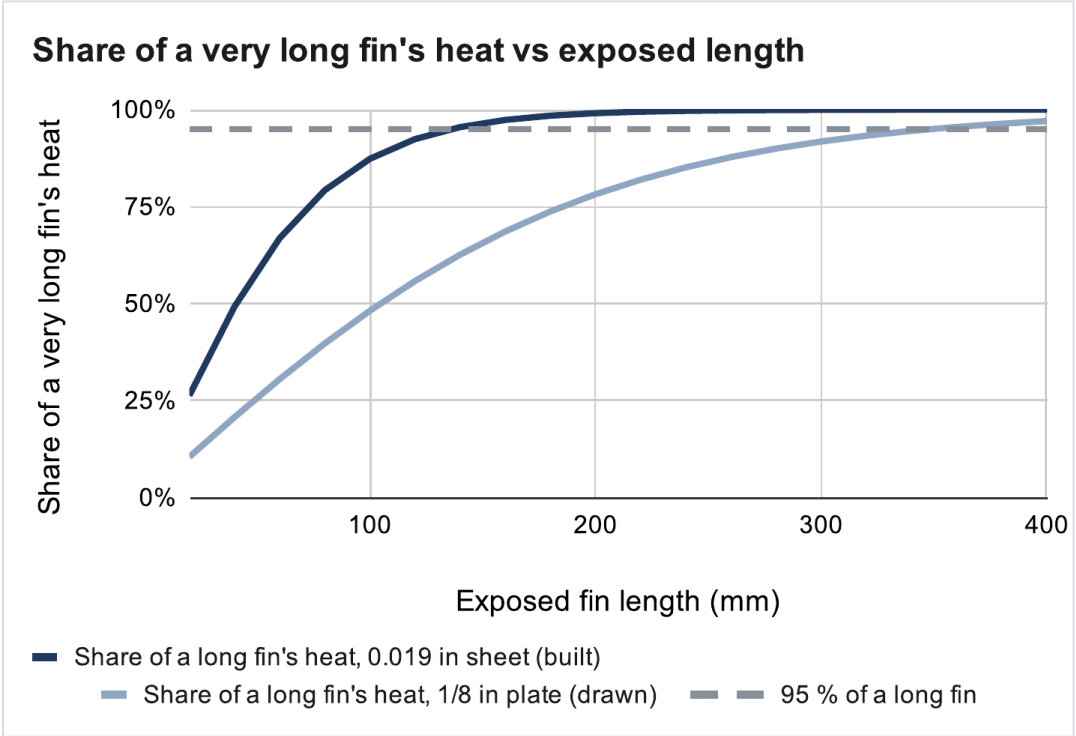
EQUATIONS

$$m = \sqrt{(2h / (k\,t))}, \quad \eta = \tanh(mL) / (mL)$$

$$95\ \%\ \text{length:}\ \ \tanh(mL) = 0.95 \rightarrow L = 1.83 / m$$

$$k = 160\ W/m{\cdot}K,\ h = 7\ W/m^2{\cdot}K,\ t = 0.48\ mm \rightarrow 136\ mm$$

Fin equations: Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3



Expected values: my calculation sheet (Google Sheets), "Heat fin" tab

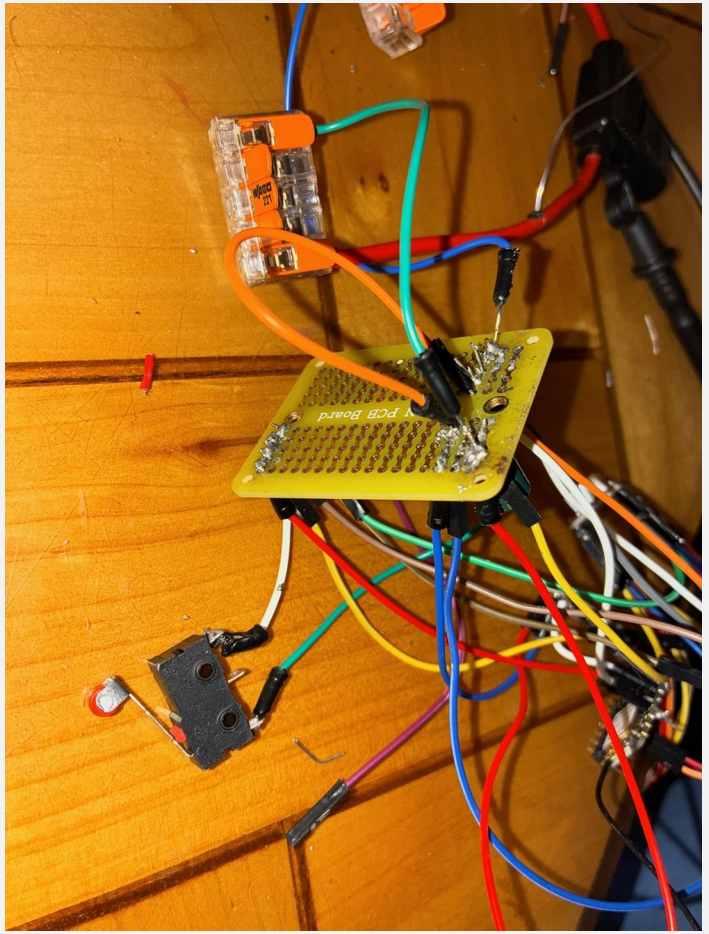
03

Homing and calibration

Why the first homing plan limited the robot, and what replaced it

First plan: bead homing

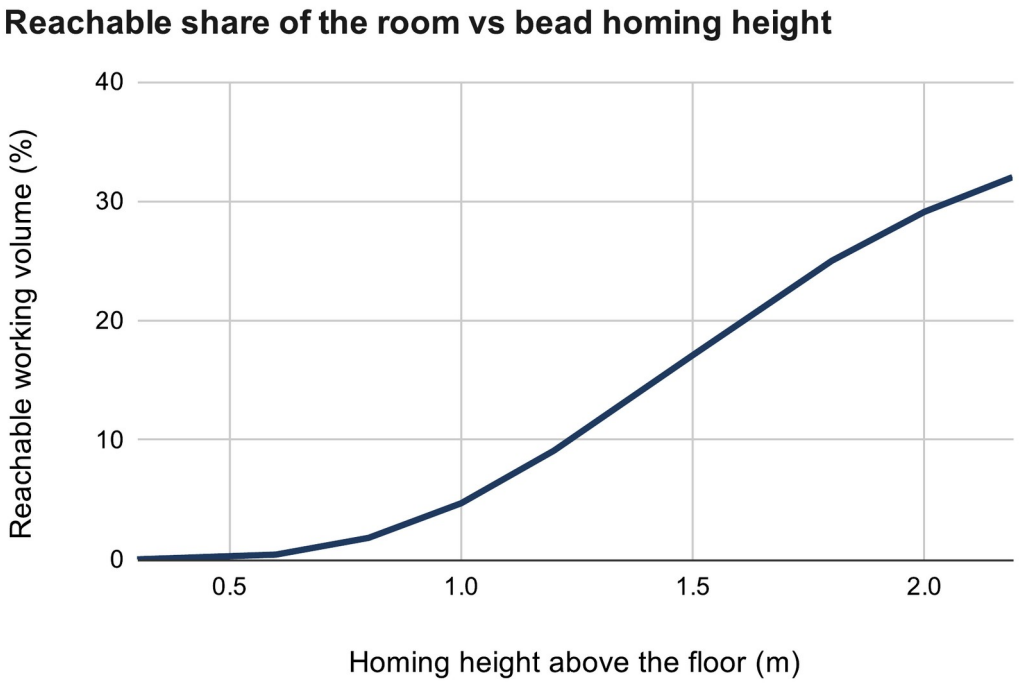
- A bead on each cable trips a switch at the outlet → a known cable length
- The robot zeroes itself from that: set it up anywhere
- I expected that to be enough, possibly even without encoders
- Detail fixed early: the switch moved below the pulley, so the bead never rides over the pulley groove



KW12-3 homing switch on the bench

The problem: beads limit the motion

- A bead cannot pass the outlet, so a cable can never be shorter than its homing length
- To home all four at one point, each bead sits at that point's length
  - then the claw can never get closer to any anchor than from that point
- Homing at the room centre 1 m up → about 5 % of the working volume reachable; 2 m up → about 29 %
- And vision alone knows cable length to ~20–25 mm, while a bead allows ~12 mm of overtravel: a bead search was not safe either



Expected values: my calculation sheet (Google Sheets), "Bead homing" tab

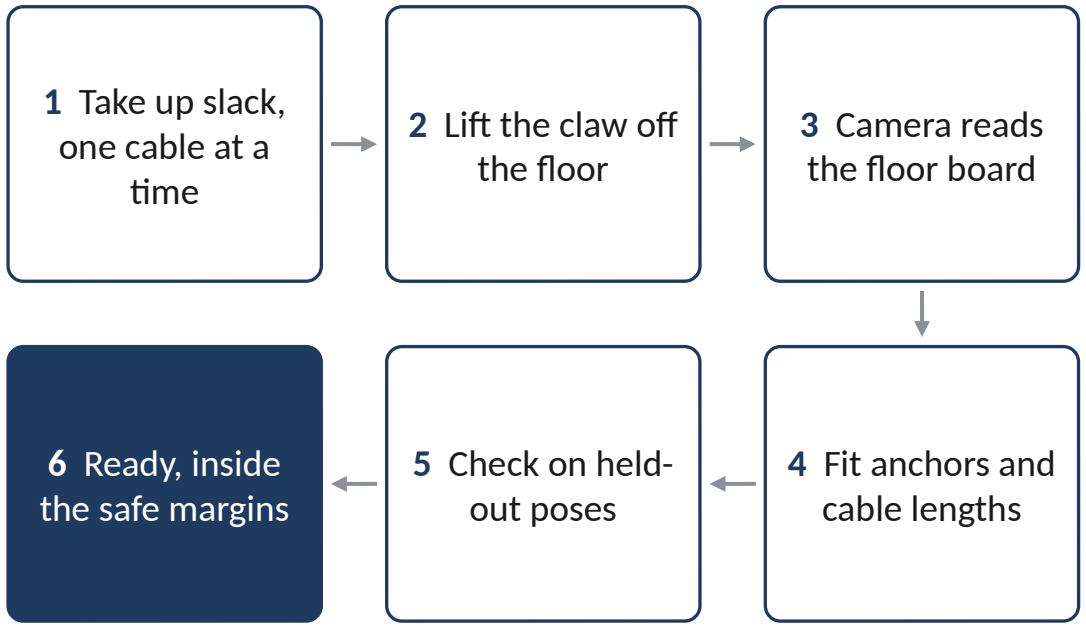
EQUATIONS

$$L_i(\mathbf{p}) \geq L_{i,\text{home}} = \|\mathbf{a}_i - \mathbf{p}_{\text{home}}\| \quad \text{for every cable } i$$

Reachable: every cable at or above its homing length

New calibration: camera, floor board and margins

- Startup: take up slack one cable at a time, lift, then look at the floor board from several poses
- Least-squares fit of the anchor positions and cable zero lengths; held-out poses check the fit
- Safety comes from margins, not precision
  - 0.4 m below the anchors and 0.2 m inside the walls unless the error bars allow more
- Beads stay as end-of-travel safety switches; encoders watch for lost steps



If the fit is not good enough: more poses, then fit again

EQUATIONS

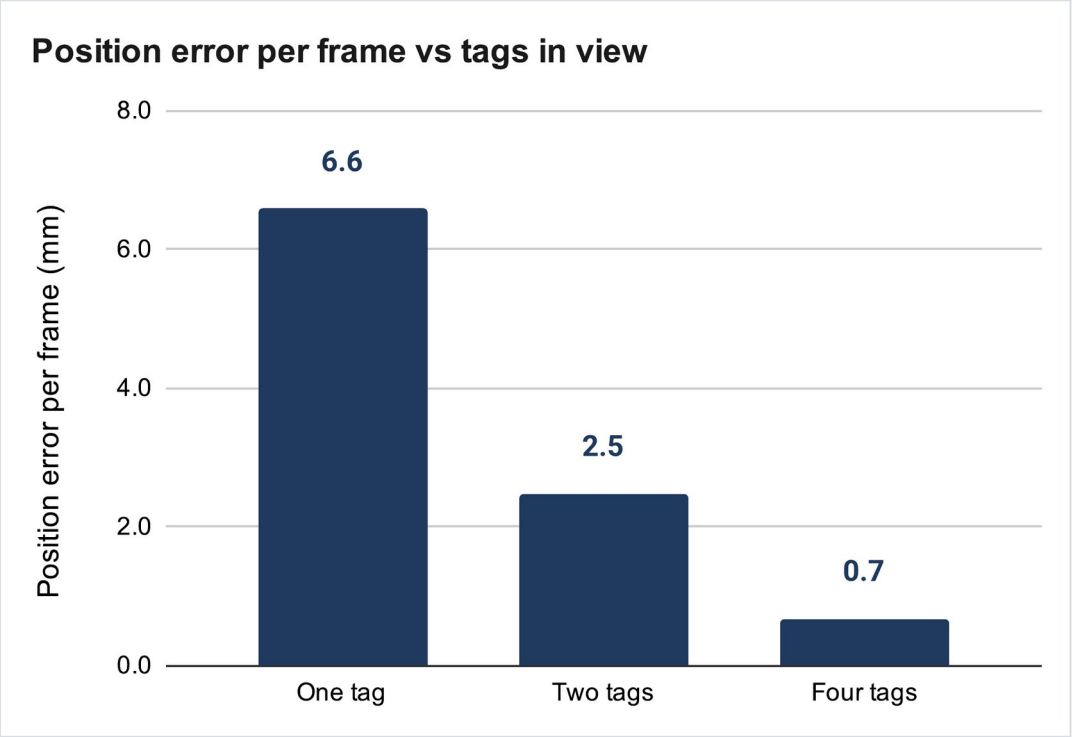
$$\min \text{ over } \mathbf{a}_i, L_{i0}: \quad \Sigma_k \Sigma_i \left(L_{i0} + d_{ik} - \|\mathbf{a}_i - \mathbf{p}_k\| \right)^2$$

$$\text{Full workspace only if } k \cdot \sigma \le \text{margin}$$

d: cable paid out at pose k (encoder); p: claw position from the camera

Why a floor board

- Angle error becomes position error through the lever arm
- One tag: 12.9 mrad × 0.51 m ≈ 6.6 mm per frame
- Four tags in view: about 0.7 mm
- The board is a printed grid at 250 mm pitch: cheap, and it works in any room
- Averaging frames helps only with random noise, not with a fixed bias



Expected values: my calculation sheet (Google Sheets), "Floor board" tab

EQUATIONS

$$\delta x = \sigma_\theta \cdot L = 12.9\ \text{mrad} \times 0.51\ \text{m} = 6.6\ \text{mm}$$

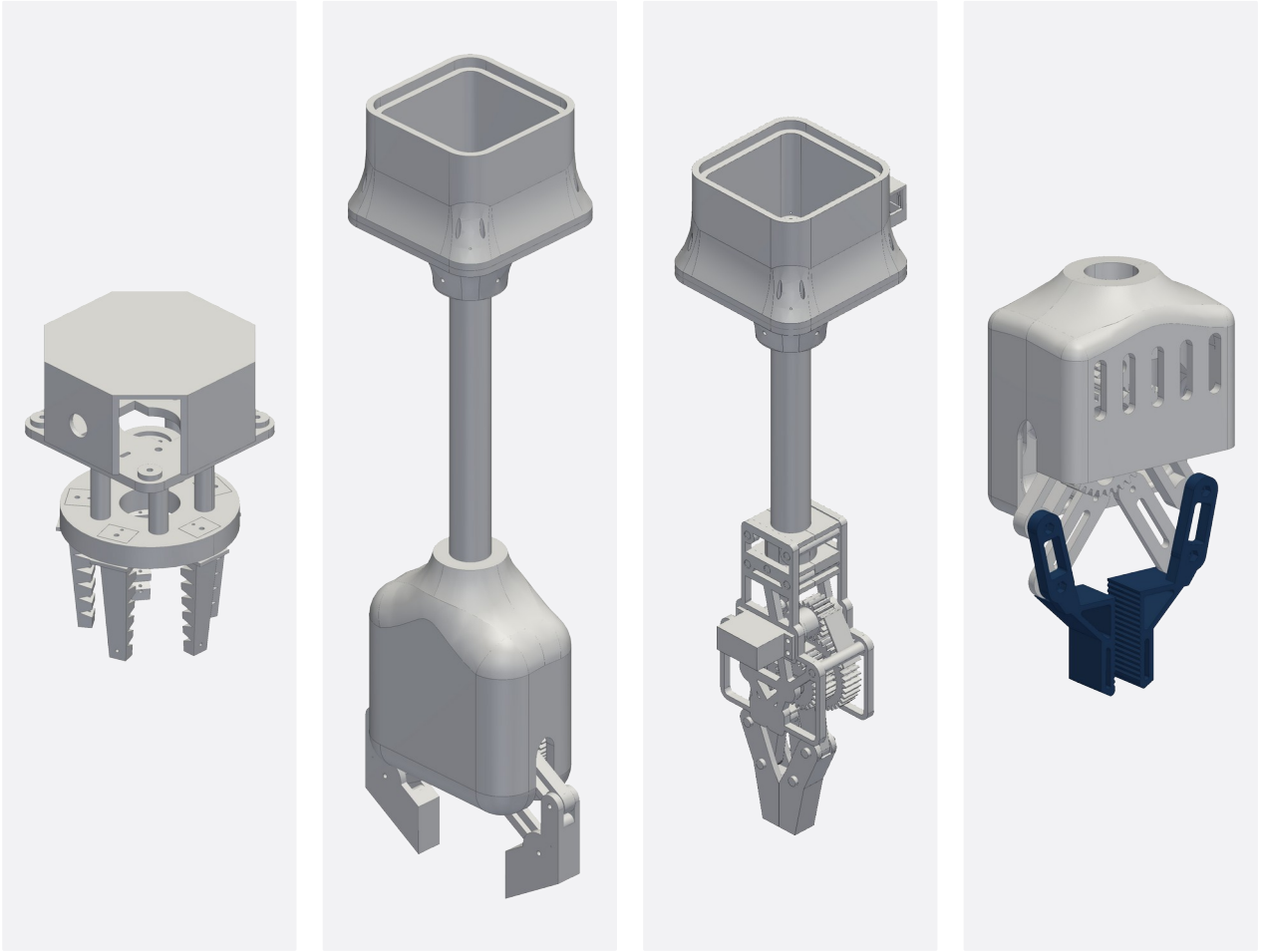
04

The claw

From fingers to a parallel-jaw gripper

Claw evolution

- Five tendon-driven TPU fingers → a geared clamp → a lighter open frame → the final parallel-jaw gripper
- Each step fixed a problem I found:
  - the first clamp's jaws could not pinch thin fabric
  - the v6 clamp was about 4.6× its mass budget
  - v7 cut the gearbox frame to 143 g
- Final: my gear housing and arms, with hands designed to fit them



Fingers, v6 clamp, v7 open frame, final gripper

Final gripper: gears and four-bar hands

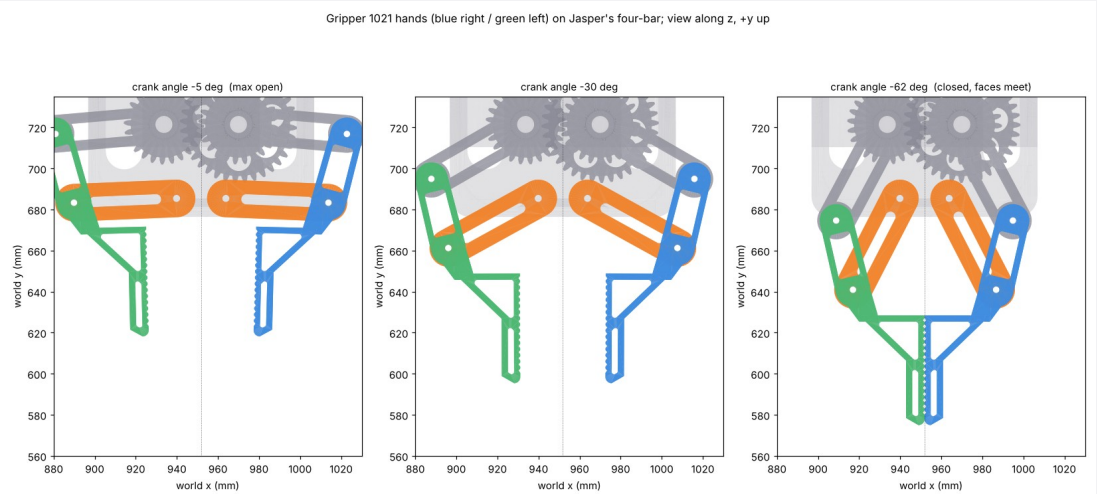
- Servo → 18T pinion → 36T/24T compound → 24T sync gear: mirror-image jaws
- Each hand: gear crank (53 mm) + swing arm (50 mm), a near-parallelogram four-bar
  - jaws stay parallel within about 1° over the stroke
- 53 mm opening; closing drops the tips ~38 mm: a scoop
- Gear teeth at servo stall: safety factor about 2
- Swept in CAD: no collisions from open to closed

EQUATIONS

Gear ratio $i = 36 / 18 = 2 \quad (\text{servo} \rightarrow \text{crank})$

Grip: $2 \mu N \geq m g \rightarrow N \geq 7.4\ \text{N per jaw} \quad (1\ \text{lb}, \mu = 0.3)$

Lewis: $\sigma = F_t / (b\, m\, Y), \quad F_t = 2\tau / d$



Hands on the linkage: open, mid, closed

First setup: an overhead camera

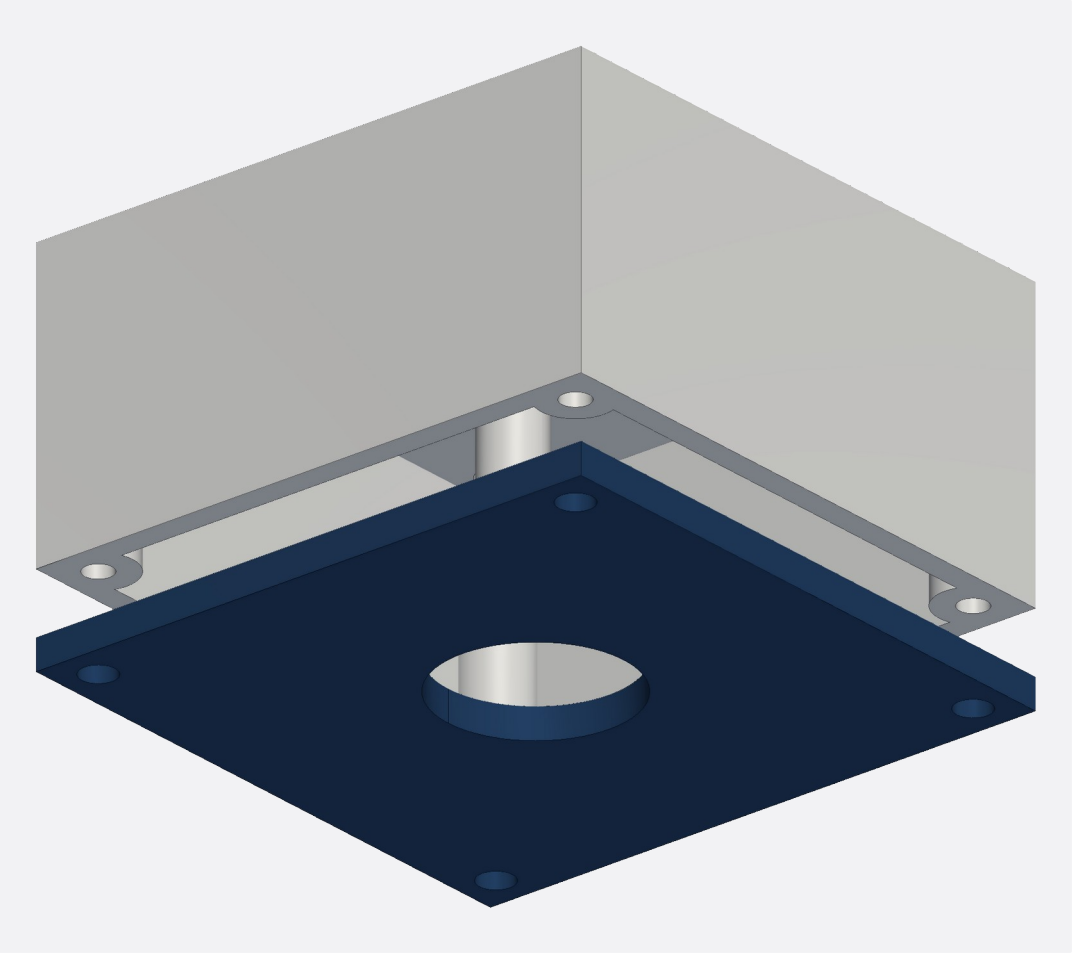
- One camera on the ceiling, looking straight down and framing the whole floor
- A detector that works from words (YOLO-World) finds “sock”, “towel”, “t-shirt” with no training
- Each item's bottom-centre pixel, where it meets the floor, becomes a floor position
  - straight-down camera: scale by the room size; tilted camera: a homography from four taped floor points
- The claw then flies to that point and grabs
- The camera board sat in a printed ceiling case: shell plus a screw-on lens bezel

EQUATIONS

Straight down: $x = u \cdot W / w_{img}, \quad y = v \cdot D / h_{img}$

Tilted: $(x, y, 1) \propto \mathbf{H} \cdot (u, v, 1), \quad \mathbf{H}$ fitted from 4 floor points

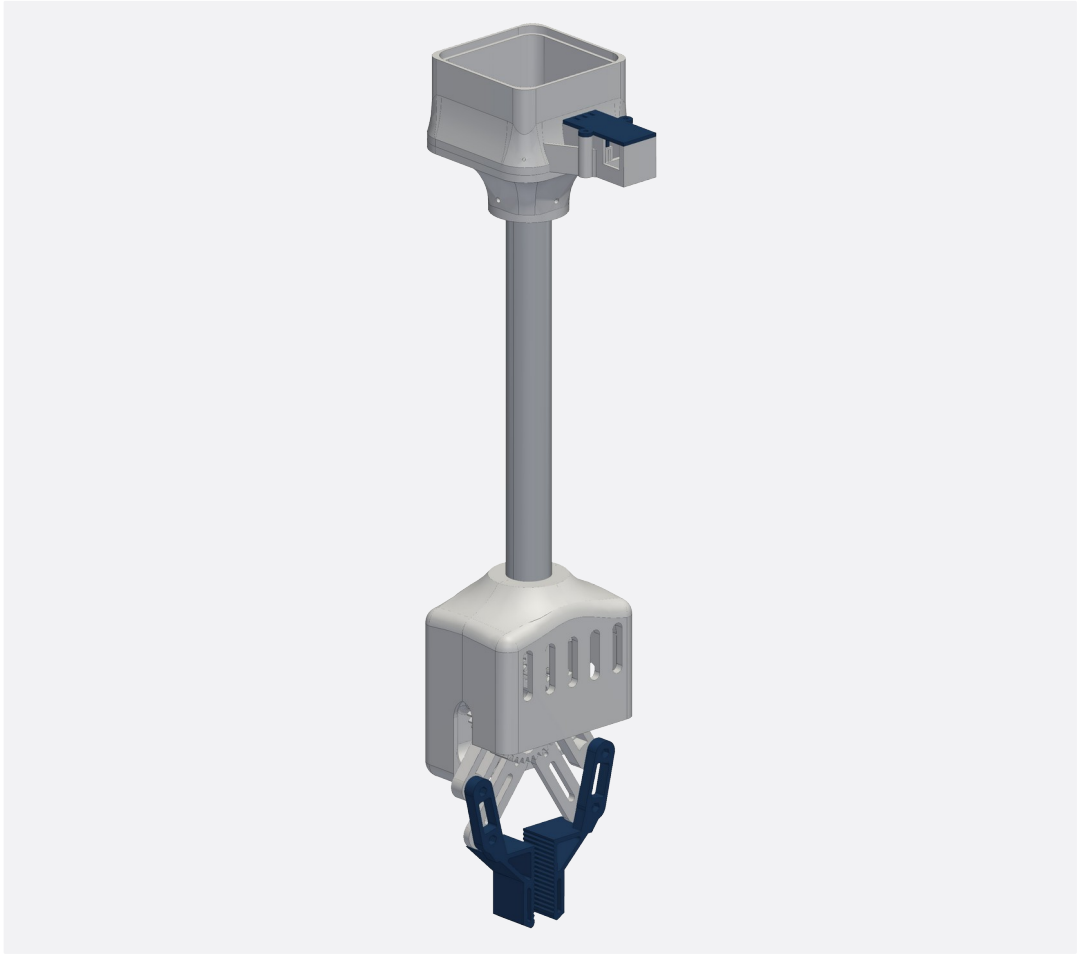
u, v: pixel; W × D: floor size; w × h: image size in pixels



Gen-1 ceiling camera case: shell and lens bezel (45 g PETG)

Moving the camera to the claw

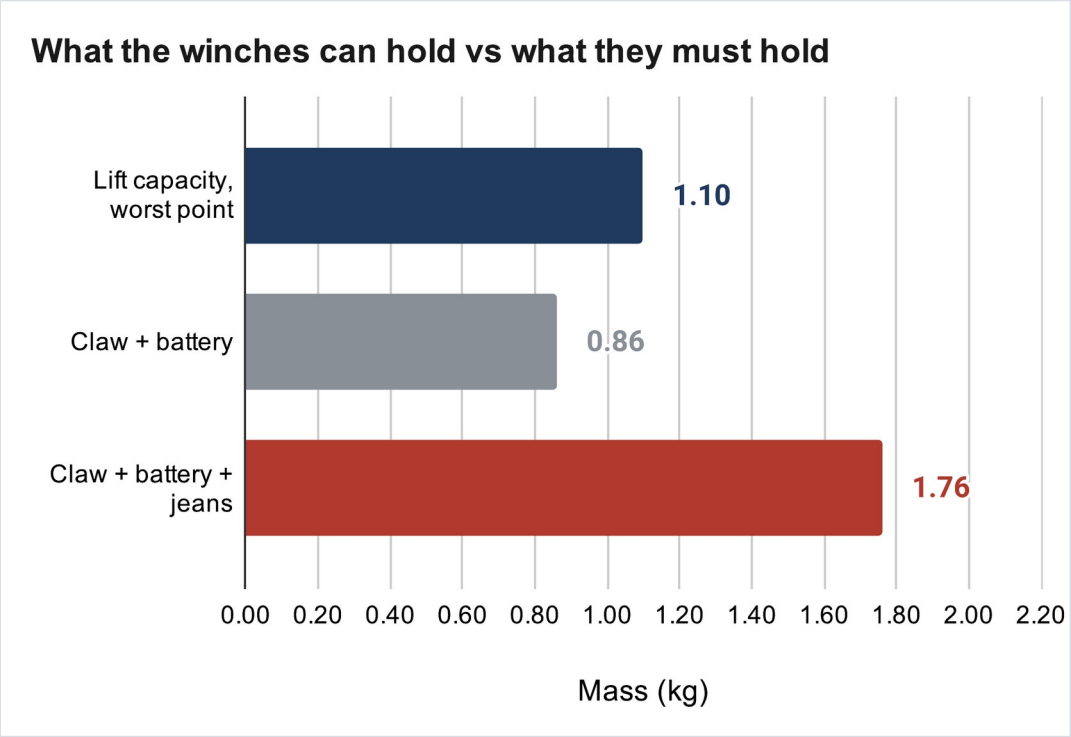
- The overhead camera was one more device to mount, wire and calibrate in every room
- The claw and cables hide the spot right where it grasps: it cannot see the grip
- Its floor map assumes a flat floor, but a pile of clothes has height
- Now: a XIAO ESP32-S3 Sense camera in a bay in the claw's housing, looking straight down the tube
- One camera, two jobs: it guides the grasp up close and reads the floor board for calibration
- Cost: it sees less of the room at once, so the robot scans by moving



Camera bay on the electronics housing (cover in navy)

Weight vs lift: where it stands

- Driver limit 1.06 A → running pull ≈ 18–24 N per winch (full vs empty spool)
  - about 1.1 kg total lift at the worst point in the room
- The claw is not weighed yet: CAD says ~0.76 kg printed, plus a 0.1 kg battery
- With jeans that is above the capacity today: this is the open problem
- Options: lighten the claw, a 3:1 winch reduction (~2.7× pull), or a lighter demo load
  - next step: a luggage-scale pull test on one station decides



Expected values: my calculation sheet (Google Sheets), "Lift budget" tab

EQUATIONS

$$\Sigma T_i u_{iz} = m\,(g + a), \quad 0 < T_i \le 11.9\ \text{N} \quad \text{(worst point in the room)}$$

$$m_{claw} \approx f\,\rho\,V + m_{tube} + m_{servo} = 0.55 \times 927\ \text{g} + 198\ \text{g} + 55\ \text{g} \approx 0.76\ \text{kg}$$

05

Manufacturing

Printed PETG parts, and a heat fin cut and drilled on a manual mill

Designing parts to be printed

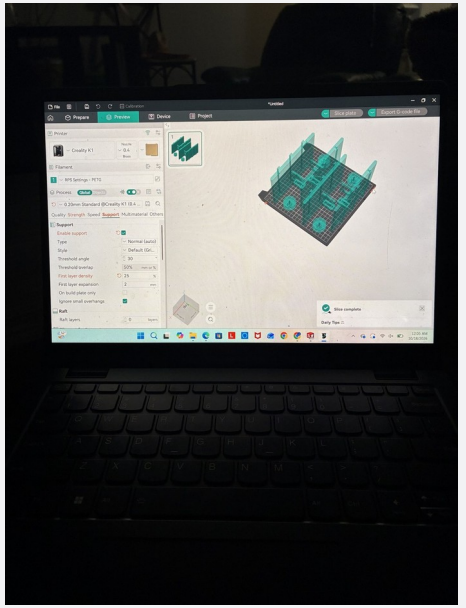
- PETG on a Creality K1 (0.4 mm nozzle, 0.2 mm layers); PLA creeps under load
- Orientation: flat face on the bed, loads along the layers, no supports on mating faces
- Printed holes come out undersize (bead rounding, shrink, first-layer squash)
  - so fits get a lead-in: 687 bearing pocket Ø14.075 with a Ø14.000 entry lip
- Heat-set M3 inserts where screws are reused
- Fewer parts: camera bay printed into the housing

EQUATIONS

Clearance $c = (D_{hole} - u) - D_{shaft}$, u = printer undersize

Spool: Ø5.2 too tight → $u \geq 0.2$ mm, so Ø5.45 gives $c \approx 0.25$ mm

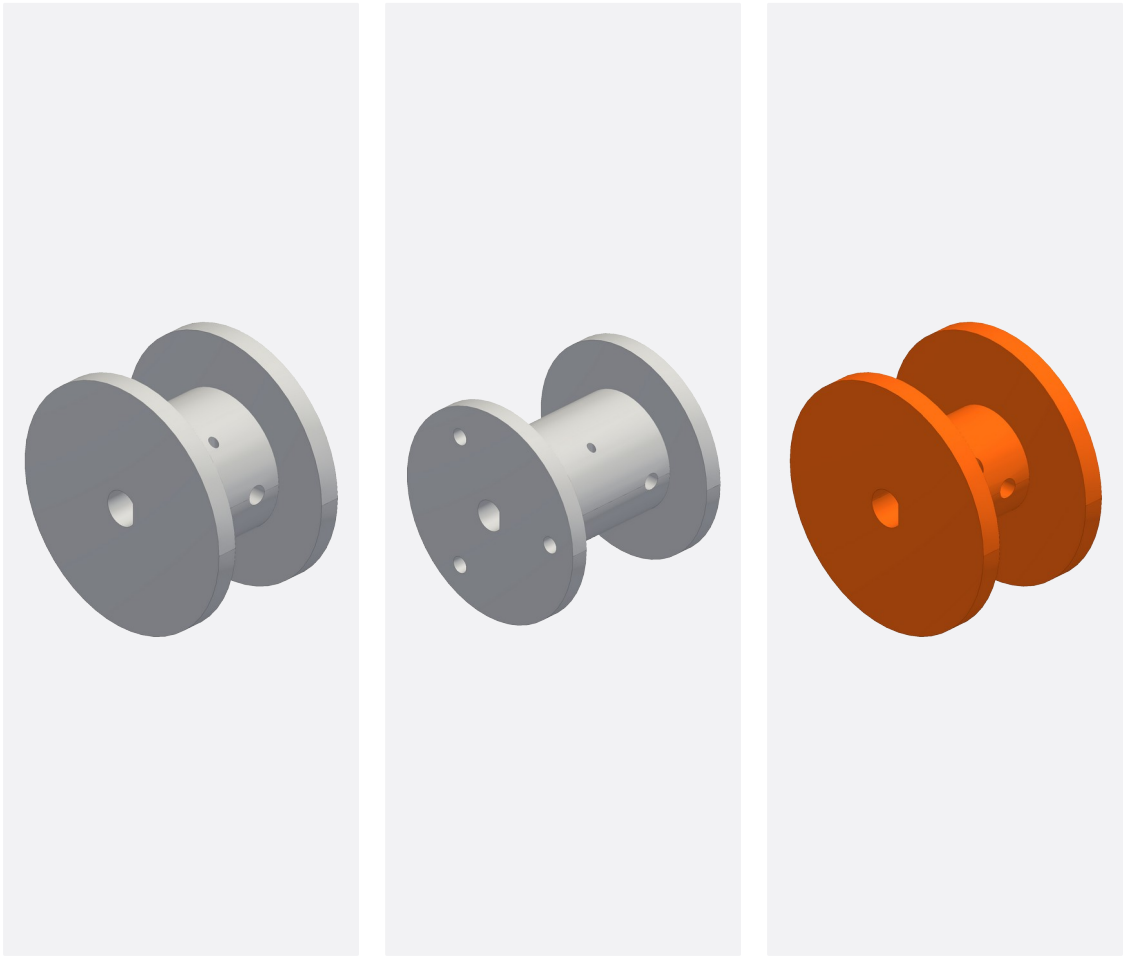




Printed PETG station shells; slicing the plate for the K1

Print, measure, adjust: the spool

- v1: longer than the motor shaft could engage, so it was shortened
- v2: the Ø5.2 bore printed too tight on the 5 mm shaft
- v3: bore opened to Ø5.45 with a D-flat; the set screw on the flat carries the torque
- Ø15 core and 16 mm drum: 20 ft of line in 5 layers with 8 mm of flange margin
- Next: a fit coupon before every new interface, measured with calipers



Spool v1, v2, v3 (CAD)

Machining the heat fin

- Shop drawing first: a 1:1 US Letter print with a hole template and inch dimensions
- Sheared the blank from 0.019 in aluminium sheet: 44 × 250 mm (±0.5)
- Manual mill, one setup: all five holes located from the bottom and left edges
  - 7/8 in pilot for the motor boss and 4 × Ø3.4 on the 31 mm motor square, ±0.1 mm to each other
- Deburred, then matte black paint except the motor contact square; thermal paste there
- Test fit on the NEMA 17: boss centred, all four screws lined up









Shear and blanks, mill setup with the drawing, drilling, the finished fin

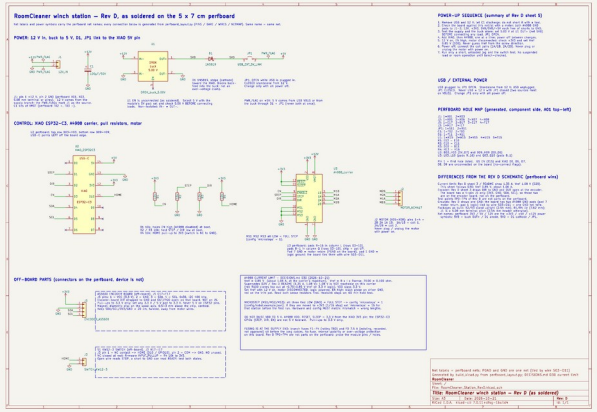
06

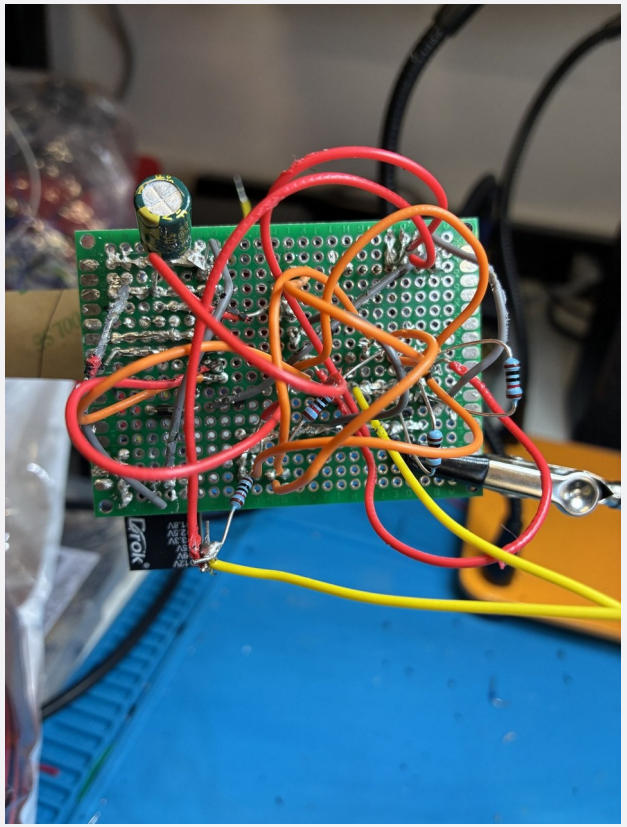
Electronics and software

The station board, a bring-up lesson, and how the robot is controlled

Station board

- The driver boots disabled (EN pull-up): a reset never steps the motor
- USB and the 5 V buck cannot back-feed each other: a link jumper plus a Schottky diode
- 100 µF at the driver's motor supply
- Connectors for the motor, homing switch, encoder and 12 V
- The KiCad schematic is generated from the soldered board's net list, so the two cannot disagree

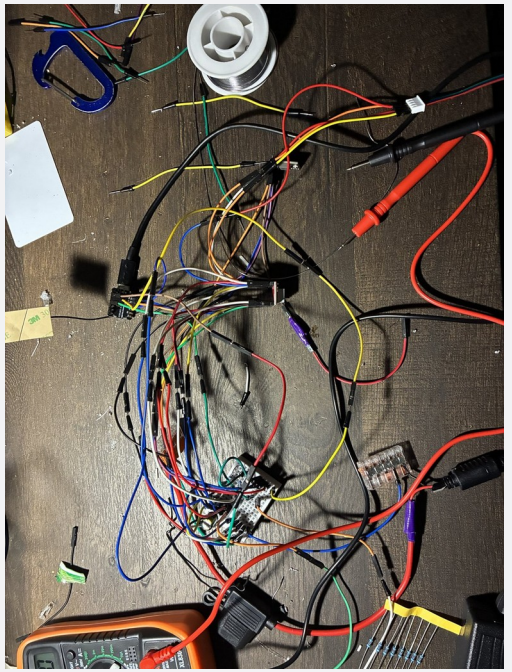




KiCad schematic (left), checked net by net against the soldered perfboard (right)

A motor that would not move

- The log said it was stepping; the shaft did not turn
- The meter said the logic rail was 1.14 V instead of 3.3 V: a bad connection
- After the fix it moved
- Since then: a meter checklist before any code
  - VDD, SLEEP, ENABLE and the buck output, every new board

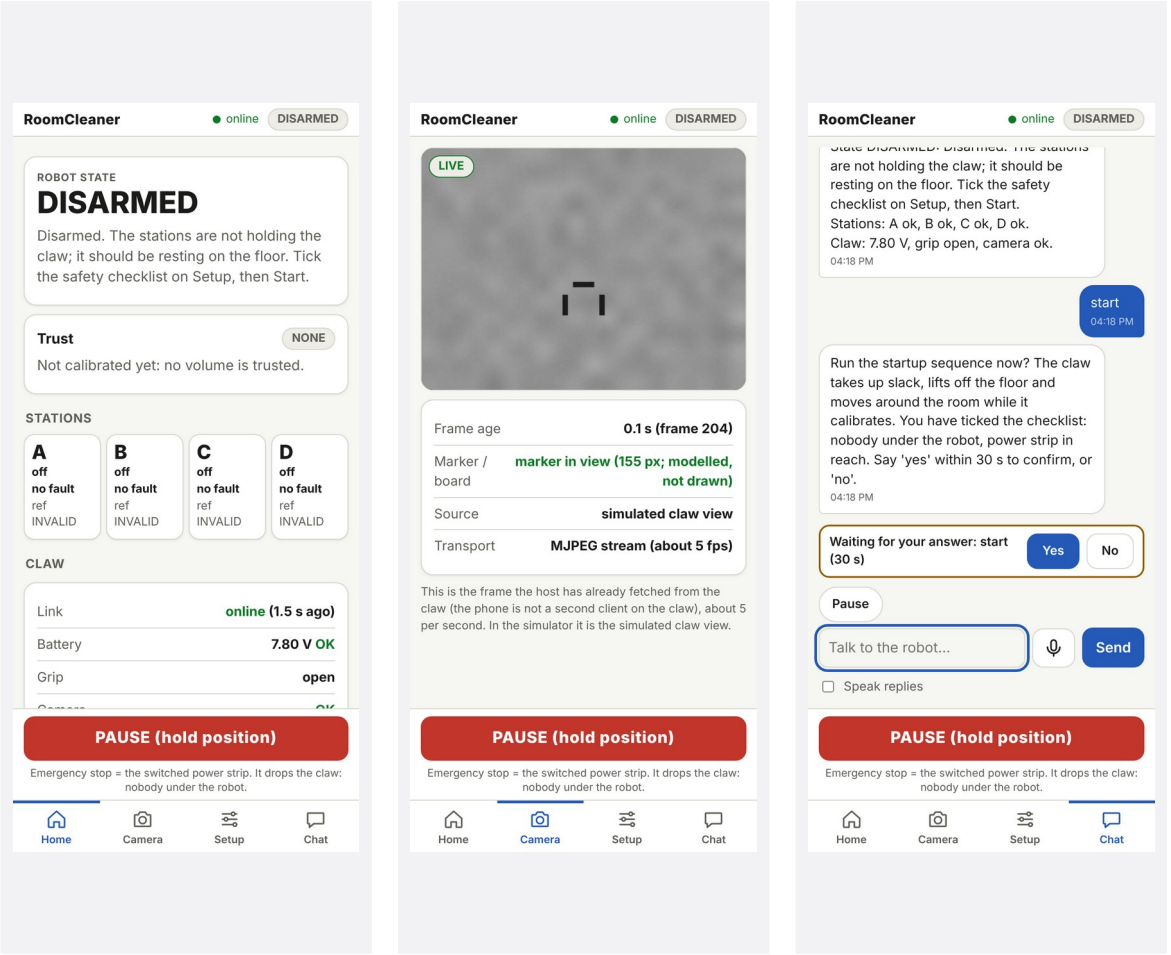




Bench wiring and bring-up

Control software and phone app

- The host plans paths; the stations execute timed segments; a lost link means hold
- Stop never disables: there is no brake, so faults and pause keep holding torque
- Phone app over home Wi-Fi: robot status, live claw camera, guided setup, chat
  - anything that moves the robot asks “yes” first; pause runs at once
- Tested in a full simulator of the stations and the claw



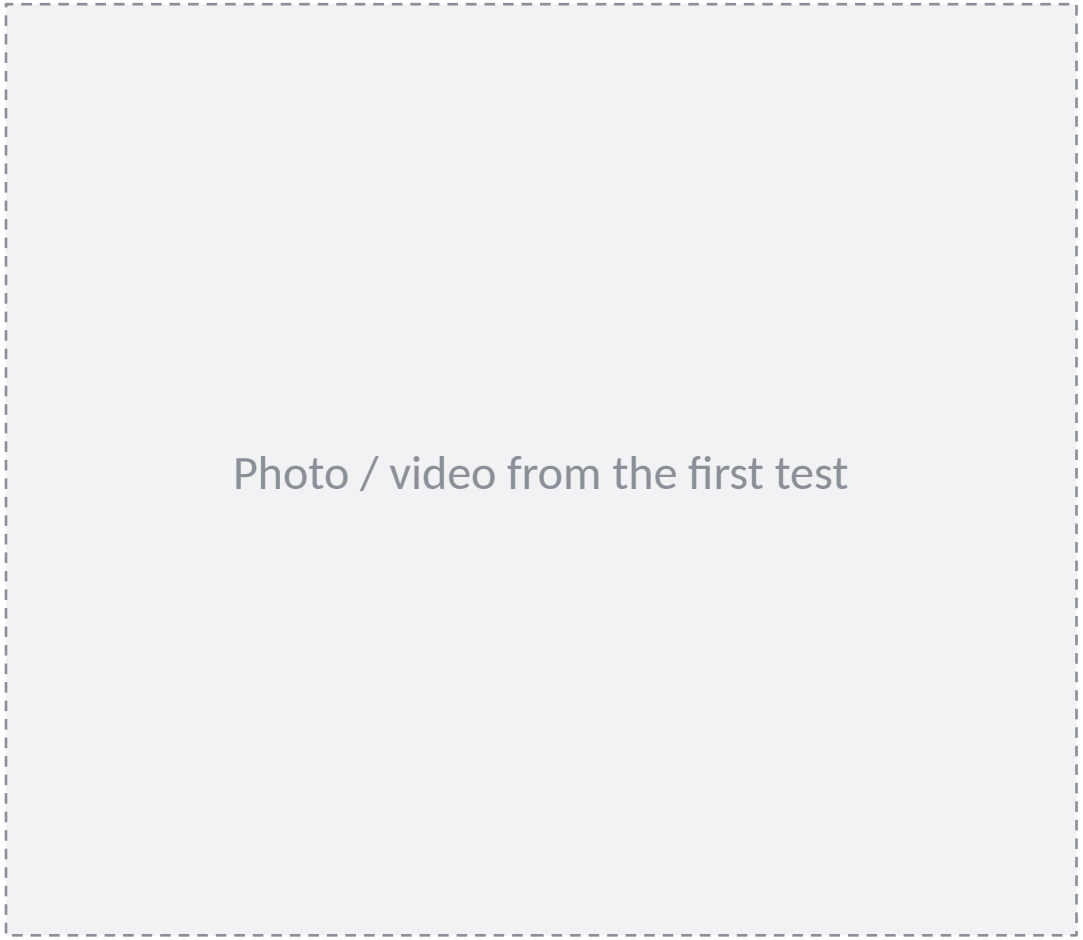
Phone app: home, camera, chat

07

Status and next steps

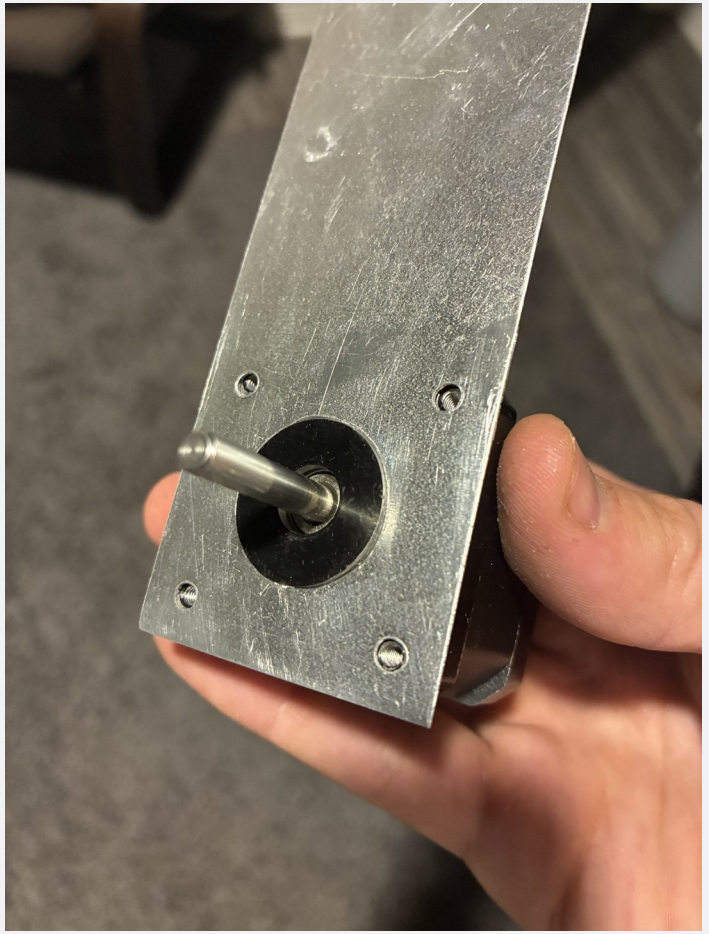
First test

- Stations up in studs, powered from the switched power strip
- Self-calibration on the real hardware: slack take-up, lift, floor-board poses, fit
- Then a short lift and a 300 mm move at 50 mm/s with the 1 lb test load
- Logged: cable lengths, encoder counts, motor temperatures and the fit's error bars
- Pass limits written before the run



Status and next steps

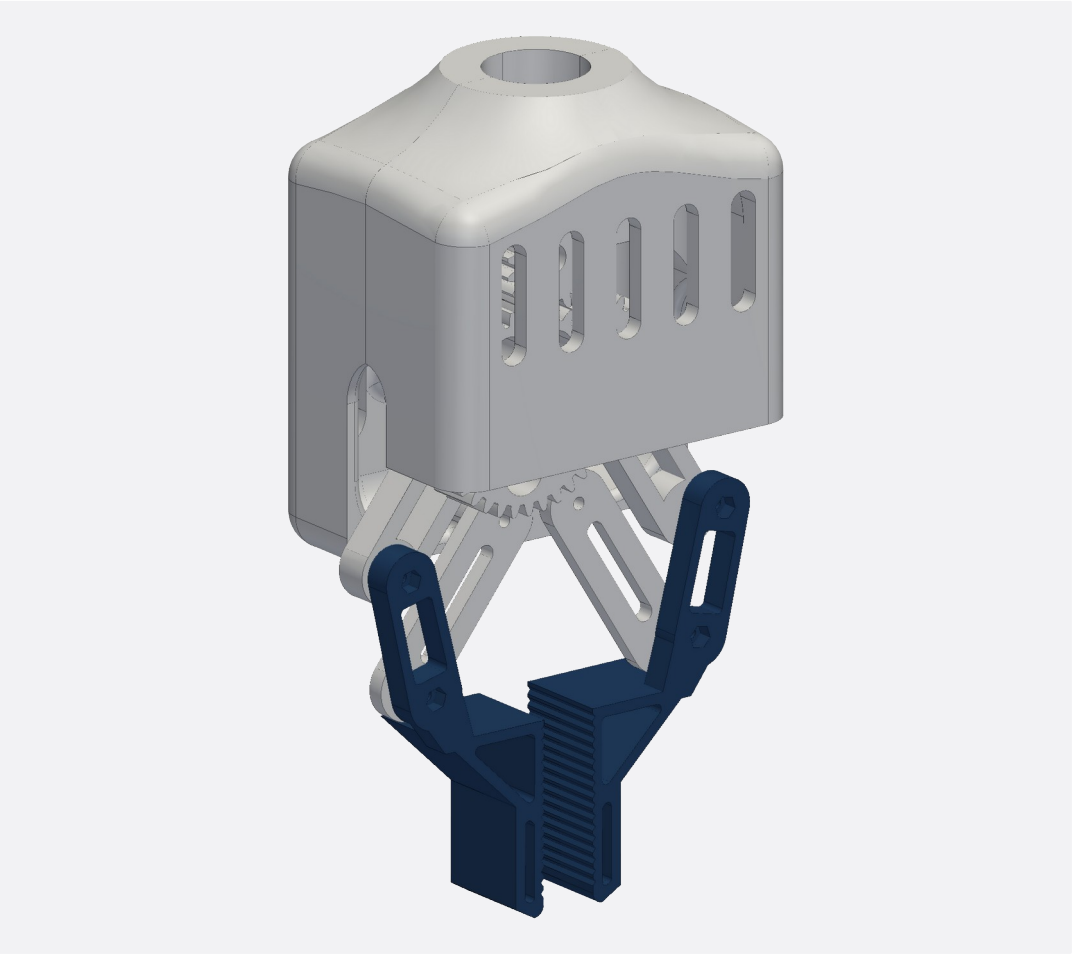
- Built: stations, station boards, heat fins, gripper parts
- Designed and checked in CAD: the hands and the camera housing
- Calibration and control: simulated, then the first test
- Next tests, with pass limits written first:
  - winch pull test (decides the capacity fix)
  - 90-min hold test: motor and printed mount under 60 °C
  - SolidWorks static and thermal studies of the station
  - four-station lift: 1 lb, 300 mm at 50 mm/s, within 2 mm per cable



Built: the heat fin test-fitted on the motor

What I learned

- Measure before trusting a margin: the driver limit and the 1.14 V rail
- Print the test coupon before the part: the spool bore
- Hold tolerance where it matters: one mill setup for the fin's holes
- Design for the worst case: stall, the corner of the room, the hottest hour
- Modularity paid for itself: every station can be pulled, fixed and put back
- Question each requirement and delete parts: the IMU, the overhead camera, the five-finger hand, the long motor harnesses

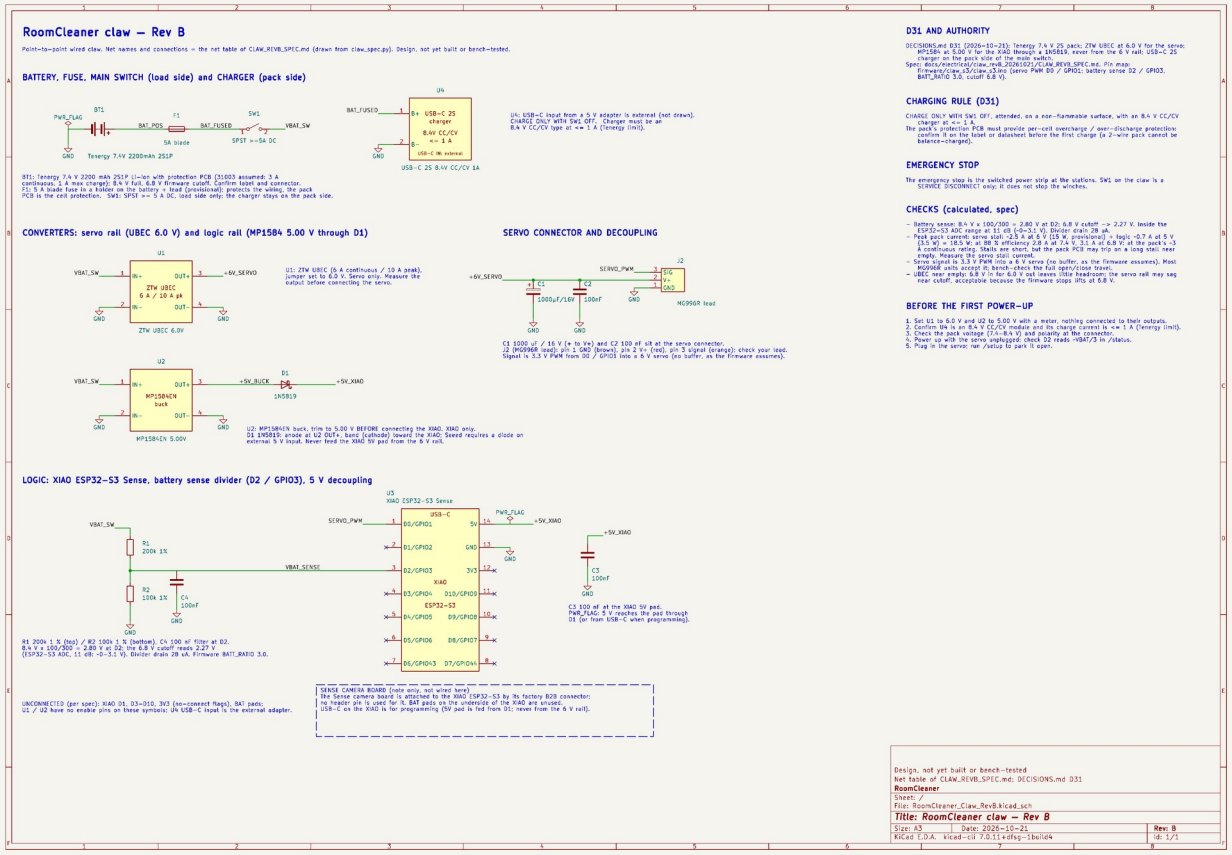


Questions

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Backup: claw electrical

- One protected 2S pack (7.4 V, 2200 mAh), 5 A fuse, main switch
- 6 V regulator for the servo only; separate 5 V buck and Schottky diode for the controller
- Battery read through 200k / 100k: 8.4 V → 2.80 V at the ADC
- Peak ≈ 2.8–3.1 A against a ≈ 3 A pack rating: stalls must stay short
- Charged on the claw only with the switch off, attended, at ≤ 1 A



Backup: gear strength (Lewis)

- Module 1.5, PETG allowable 22.5 MPa
- At 6 V servo stall: pinion SF 2.05, 36T gear 2.50, sync gear 2.00 (one jaw carrying the load)
- Module 1 was screened first: SF 0.35–0.41, so the module went up





EQUATIONS

$$\sigma = F_t / (b\, m\, Y),\quad F_t = 2\tau / d,\quad SF = \sigma_{allow} / \sigma$$

Backup: safety

- One hard stop: the switched power strip; faults and pause hold torque
- Stations anchor into studs; the line load is mostly pull-out and prying
- Nobody stands under the robot while it moves
- Claw battery: protected pack, charged with the switch off and attended
- No safety-rated function is claimed

Backup: calculations and sources

- Every chart comes from my calculation sheet in Google Sheets, one tab per topic
  - cable tension, winch pull, heat fin, motor temperature, wiring, bead homing, floor board, lift budget, gears
- Every result is a formula on its inputs; inputs are marked, with their source or assumption
- References used:
  - Incropera, DeWitt, Bergman, Lavine: Fundamentals of Heat and Mass Transfer, ch. 3 (extended surfaces)
  - Allegro MicroSystems: A4988 datasheet (current limit)

RoomCleaner engineering calculations (Google Sheets)